

# Oklo Inc

## Next-Generation Fast Fission Microreactors — Sam Altman's AI Nuclear Bet

*PhD Due Diligence Report | BLRI-DD-OKLO-AUG2026*

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*Six-Method Framework: Quantitative · Qualitative · Ethnographic · Superforecasting · Adversarial-Collaborative · Seldon Thesis*

*RAG Corpus: EDGAR 139,756ch · WSJ 335,496ch · FT 109,507ch · NIKKEI 397,454ch · FA 148,327ch · ST\_PHD 60,453ch*

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### PART I — SELDON THESIS AXIS

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#### Aurora Microreactor: 1.5MW to 15MW Firm Carbon-Free Power at the Grid Edge

Oklo Inc is developing the Aurora fast fission microreactor — a liquid metal-cooled reactor using HALEU (High-Assay Low-Enriched Uranium) fuel capable of producing 1.5 to 15 MW of electrical power. Unlike large nuclear plants requiring 10+ years to build, Aurora is designed for factory fabrication and modular deployment: a single Aurora unit can be deployed at a remote mining site, military base, AI datacentre, or industrial facility in 2-3 years. OpenAI CEO Sam Altman chairs Oklo's board, providing the company direct access to the largest AI compute demand signal in the world. In Seldon K-variable terms: if Oklo can commercialise microreactors, it decentralises energy production ( $\xi$  uplift) and powers AI compute at the grid edge.

Oklo received the first combined licence application (COLA) acceptance from the NRC for an advanced non-light water reactor in 2021. While the original licence was denied in 2022 (insufficient safety data), Oklo refiled in 2023 with an expanded application. The technology is real; the question is regulatory timeline and commercial deployment cadence.

*Source: moomoo OpenD [2026-08-13] · Oklo NRC COLA filings · DOE HALEU programme*

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### PART II — QUALITATIVE ANALYSIS

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#### HALEU Fuel: The Critical Path Dependency

Aurora requires HALEU (19.75% U-235 enriched uranium) — not commercially available at scale today. The DOE HALEU consortium (Centrus, URENCO) is developing domestic supply, but production capacity is

limited. Oklo has a HALEU supply agreement with the DOE; however, commercial supply chain buildout is the critical path risk to Aurora deployment timelines.

### Sam Altman / OpenAI Connection: Distribution Before Revenue

Sam Altman's chairmanship represents both a distribution advantage (direct AI demand signal) and a reputational risk (Altman's other commitments at OpenAI). Oklo signed a 20-year power purchase agreement with Switch, a major datacentre operator, for up to 12 GW of nuclear capacity — a remarkable commercial pipeline for a pre-revenue company with no deployed reactors.

Source: Oklo S-1 SEC filing · DOE HALEU programme status · Switch PPA announcement

## PART III — QUANTITATIVE ANALYSIS

**VALUATION SNAPSHOT [moomoo OpenD 2026-08-13]**  
Price: \$46.110 | Prev Close: \$45.130 | Mkt Cap: \$8.577B  
EPS TTM: -\$0.720 | Net Income: -\$133.9M | Div Yield: 0%  
52w: \$36.610–\$193.840 | PE TTM: -49.05x | Shares: 186.0M

| Metric             | Value              | Source       |
|--------------------|--------------------|--------------|
| Price (2026-08-13) | \$46.110           | moomoo OpenD |
| Market Cap         | \$8.577B           | moomoo OpenD |
| EPS TTM            | -\$0.720           | moomoo OpenD |
| Net Income TTM     | -\$133.9M          | moomoo OpenD |
| 52-Week Range      | \$36.610–\$193.840 | moomoo OpenD |
| Div Yield          | 0%                 | moomoo OpenD |

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## PART IV — COMPETITIVE LANDSCAPE

| Competitor             | Type                  | Mkt Cap  | vs OKLO                                                             |
|------------------------|-----------------------|----------|---------------------------------------------------------------------|
| NuScale Power (SMR)    | SMR (light water)     | \$4.059B | SMR NRC-licensed; OKLO advanced fast fission; OKLO smaller MW range |
| Kairos Power (private) | Fluoride salt reactor | Private  | DOE demo funded; OKLO has Aurora filing advantage                   |
| TerraPower (private)   | Natrium sodium-cooled | Private  | Gates-backed; OKLO microreactor smaller scale niche                 |
| X-energy (private)     | Pebble bed HTGR       | Private  | OKLO listed; all others private; OKLO capital access advantage      |

## PART V — ETHNOGRAPHIC PORTRAITS

### FINDING F-5-A: Portrait 1: The Remote Mining Operation Manager (COMPOSITE)

An operations manager at a Canadian copper mine in Northern Ontario evaluates Aurora for grid-independent power. 'Our current diesel costs \$0.42/kWh. An Aurora at \$0.15/kWh would halve our energy bill and eliminate fuel trucking risk. We'd sign a 20-year PPA tomorrow.' Represents the off-grid industrial market — the most immediately addressable market for microreactors where power price is already very high.

*COMPOSITE / ILLUSTRATIVE — drawn from remote industrial energy patterns*

### FINDING F-5-B: Portrait 2: The US Air Force Base Energy Security Officer (COMPOSITE)

A US Air Force energy security director at a classified installation notes the Aurora appeal: 'Our base needs resilient power independent of the civilian grid. Aurora is the only sub-10MW nuclear solution approaching NRC licensure. If it deploys on schedule, it's a national security asset.' Represents the US military market — where energy resilience is a mission-critical requirement and cost is secondary.

*COMPOSITE / ILLUSTRATIVE — drawn from US military energy security patterns*

### FINDING F-5-C: Portrait 3: The Speculative Tech Investor (COMPOSITE)

A tech investor who backed Sam Altman's early OpenAI funding holds Oklo shares. 'Altman's track record at OpenAI means I give Oklo technological vision credibility. The \$8.6B market cap for a pre-revenue nuclear company is speculative, but the Switch 12 GW PPA gives me a demand anchor.' Represents the retail/tech-adjacent investor base that drives OKLO's speculative premium — the 52-week range (\$36.61–\$193.84) reflects extreme sentiment volatility.

*COMPOSITE / ILLUSTRATIVE — drawn from speculative tech investor patterns*

## PART VI — ADVERSARIAL COLLABORATION (5 HYPOTHESES)

### HH1: HH1: NRC licensing denied again — OKLO grounded for 3+ more years

#### EVIDENCE FOR:

NRC denied OKLO's first COLA in 2022; second application may face same fate if safety analysis framework is insufficient for novel fast fission design; no other licensed fast reactor design exists in the US

#### EVIDENCE AGAINST:

Oklo hired former NRC commissioners and staff; second COLA incorporated all 2022 deficiency corrections; NRC's Advanced Reactor Program (ARC) is explicitly facilitating novel reactor designs

#### QUANTITATIVE TEST

$P(\text{COLA denial again before 2027})=25\%$

#### VERDICT: BEAR — NRC denial would be existential for the current deployment timeline

Implication: Monitor NRC COLA review status; track Oklo's safety analysis submissions quarterly

## HH2: HALEU supply chain fails to materialise

### EVIDENCE FOR:

HALEU production in the US is limited to DOE Centrus demonstration (1.8 MT/year for 3 years); no commercial-scale HALEU enrichment capacity exists; Aurora needs commercial supply by 2027

### EVIDENCE AGAINST:

DOE has prioritised HALEU as national security infrastructure; URENCO US and Centrus have expansion plans; Oklo's DOE supply agreement covers initial units

### QUANTITATIVE TEST

P(HALEU commercial supply insufficient for Aurora deployment by 2028)=35%

### VERDICT: BEAR — HALEU bottleneck is the highest-probability deployment risk

Implication: Monitor DOE HALEU programme milestones; Centrus enrichment capacity expansion announcements

## HH3: HH3: Switch PPA represents real demand — OKLO is undervalued

### EVIDENCE FOR:

Switch datacentre company signed PPA for up to 12 GW of Aurora capacity; at \$0.15/kWh and 12 GW × 8,760 hours = \$16B/year potential revenue; current \$8.6B mkt cap prices zero of this

### EVIDENCE AGAINST:

12 GW is aspirational — Aurora has never been built; Switch PPA is contingent on successful deployment; timeline is 2027-2035+

### QUANTITATIVE TEST

P(Switch PPA generating material revenue by 2029)=15%

### VERDICT: BULL — if commercial deployment begins, OKLO re-rates dramatically

Implication: Monitor NRC milestone; Switch relationship announcements; Aurora pilot contract

## HH4: HH4: Sam Altman departure destroys OKLO narrative premium

### EVIDENCE FOR:

Altman's charismatic involvement is a large component of OKLO's \$8.6B valuation premium over fundamental cash flows; any Altman-related controversy at OpenAI could cascade to OKLO

### EVIDENCE AGAINST:

Altman's governance role at OKLO is independent of OpenAI; OKLO has independent management team; commercial fundamentals (Switch PPA) are real

### QUANTITATIVE TEST

P(Altman departure causing >30% OKLO share decline)=15%

**VERDICT: BEAR — key man risk is concentrated and narrative-driven**

Implication: Monitor Altman's OpenAI governance situation; OKLO board independence

## HH5: HH5: DOE loan guarantee unlocks OKLO's balance sheet

### EVIDENCE FOR:

DOE Loan Programs Office (LPO) is actively guaranteeing advanced nuclear projects; a \$2-3B DOE loan guarantee would fund Aurora commercial deployment without equity dilution

### EVIDENCE AGAINST:

LPO has already committed to TerraPower (Natrium) and Kairos; OKLO's HALEU sourcing from DOE strengthens relationship

### QUANTITATIVE TEST

P(DOE loan guarantee >\$1B for OKLO by 2026)=30%

**VERDICT: BULL — DOE backstop would de-risk deployment timeline**

Implication: Monitor LPO application status; DOE nuclear policy announcements

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## PART VII — SUPERFORECASTING (10 SCENARIOS)

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### SCENARIO 1: NRC COLA approved — Aurora enters construction (P = 22%%)

NRC approves OKLO's combined licence application; first Aurora unit enters construction at Idaho National Laboratory; stock re-rates to \$80+

Driver: NRC COLA review status; safety analysis submissions

Falsifier: NRC requests additional information; review extends past 2026

### SCENARIO 2: HALEU supply secured from DOE/Centrus for first units (P = 35%%)

DOE HALEU supply chain delivers sufficient material for 3-5 Aurora units; deployment pipeline secured through 2028; market validates commercial path

Driver: DOE HALEU programme milestone announcements

Falsifier: Centrus capacity insufficient; DOE reprioritises HALEU for defence

### SCENARIO 3: Base case: regulatory progress, speculative hold \$40-60 (P = 40%%)

NRC review continues; OKLO maintains HALEU supply agreement; Switch PPA intact; no material revenue; stock volatile between \$40-60

**Driver:** Quarterly NRC milestone updates; OKLO cash position

**Falsifier:** Any regulatory setback triggers significant selloff

#### **SCENARIO 4: Altman deploys OpenAI compute on Aurora power (P = 12%%)**

OpenAI announces plan to power AI training cluster with Aurora microreactors; direct Altman ecosystem integration; stock >\$100

**Driver:** OpenAI energy strategy announcements; Altman public statements

**Falsifier:** OpenAI selects alternative power source; nuclear economics unfavourable vs. grid

#### **SCENARIO 5: Stock retests \$36 low on deployment delay (P = 30%%)**

NRC requests additional information; Aurora deployment delayed to 2029; Switch PPA renegotiated; cash burn concern; stock below \$40

**Driver:** Two consecutive NRC information request letters; HALEU delay

**Falsifier:** NRC approval + DOE loan guarantee announced simultaneously

#### **SCENARIO 6: OKLO acquired by utility or defence contractor (P = 10%%)**

Dominion, Duke Energy, or L3Harris acquires OKLO for advanced reactor IP + NRC relationship; premium 50% to market price

**Driver:** Nuclear utility M&A activity; defence energy security contract

**Falsifier:** CFIUS review delays; acquirer withdraws on regulatory risk

#### **SCENARIO 7: DOE national security deployment for military bases (P = 18%%)**

Pentagon prioritises OKLO Aurora for forward operating base power; classified contract; national security funding bypasses commercial NRC timeline

**Driver:** DoD energy security budget; classified procurement announcements

**Falsifier:** Military selects alternative energy technology; solar+battery preferred

#### **SCENARIO 8: Mining sector drives commercial demand (P = 20%%)**

Rio Tinto or Barrick signs Aurora PPA for remote mine power; proves commercial viability beyond datacentres; first non-utility customer

**Driver:** Mining company energy security announcements; off-grid power procurement

**Falsifier:** Mining companies opt for solar+battery+diesel hybrid vs. nuclear

**SCENARIO 9: HALEU shortage triggers pivot to LEU+ fuel design (P = 15%%)**

OKLO redesigns Aurora to use LEU+ (< 10% enriched, more available) to circumvent HALEU bottleneck; engineering pivot buys time

**Driver:** OKLO engineering announcements; fuel design patent filings

**Falsifier:** Fuel redesign extends licensing timeline by 3+ years

**SCENARIO 10: Bull case: \$5B+ revenue pipeline by 2030 — stock \$120 (P = 8%%)**

NRC approval + DOE funding + Switch PPA Phase 1 delivery; first Aurora operational; \$200M+ revenue by 2029; re-rates to \$100+

**Driver:** All three milestones firing; licence + funding + first delivery

**Falsifier:** NRC delays + HALEU shortage + funding gap = restructuring

**PART VIII — SELDON VARIABLE DEEP DIVE**

| Variable | OKLO Impact                                                  | Direction   | Magnitude             |
|----------|--------------------------------------------------------------|-------------|-----------------------|
| K        | Decentralised AI compute power via microreactor              | ↑ Long-term | K+ if commercialised  |
| ξ        | Grid-edge nuclear = energy access for remote communities     | ↑ Long-term | ξ+ on decade horizon  |
| Ω        | US nuclear sovereignty reduces energy supply dependence      | ↓ Reduces   | Ω- mild (pre-revenue) |
| I        | DOE/NRC relationship validates US advanced reactor programme | ↑ Mild      | I+ moderate           |

**PART IX — RISK REGISTER**

| Risk                       | Probability     | Impact   | Mitigation                                                    |
|----------------------------|-----------------|----------|---------------------------------------------------------------|
| NRC COLA denial            | MEDIUM (25%)    | CRITICAL | Strengthened application; former NRC staff hired              |
| HALEU supply chain failure | HIGH (35%)      | CRITICAL | DOE supply agreement; Centrus partnership                     |
| Key man risk (Altman)      | LOW (15%)       | HIGH     | Independent management; commercial PPA anchor                 |
| Pre-revenue cash burn      | HIGH (ongoing)  | HIGH     | DOE loan guarantee pursuit; SPAC capital runway               |
| Technology not yet proven  | HIGH (inherent) | CRITICAL | Fast fission physics validated; engineering challenge remains |

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## CONCLUSION — 6-METHOD SYNTHESIS VERDICT

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**CONVICTION VERDICT: HIGHLY SPECULATIVE — REAL TECHNOLOGY, PRE-REVENUE, HIGH BINARY RISK**

Oklo at \$46.110 (mkt cap \$8.577B) is the highest-conviction long-duration technology bet in the energy sector. The Aurora microreactor addresses a real market (off-grid, datacentre, military power) with a differentiated physics approach (fast fission, HALEU). Sam Altman's involvement validates the AI demand signal. Key risks: NRC licensing timeline and HALEU supply. Seldon alignment: ξ+ long-duration (decentralised energy). HUMAN\_ONLY\_MANUAL. All figures: moomoo OpenD [2026-08-13]. RAG Corpus: EDGAR 139,756ch · WSJ 335,496ch · FT 109,507ch · NIKKEI 397,454ch · FA 148,327ch · ST\_PHD 60,555ch.